Assignment 4

October 15, 2026

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Roll No. 135

0.1 Gradient Descent Algorithm

find the local minima of the function $y=(x+3)^2$ starting from the point $x=2$

[1]:
```
current_x = 2
rate = 0.01 # Learning rate
precision = 0.000001  # This tells us when to stop the algorithm
delta_x = 1
max_iterations = 10000 # Maximum number of iterations
iteration_counter = 0
```

[2]:
```
# dy/dx of eqn = 2*(x+3)
def slope(x):
    return 2*(x+3)

def value_y(x):
    return (x+3)**2
y = []
x = []
y.append(value_y(current_x))
x.append(current_x)
```

[3]:
```
while delta_x > precision and iteration_counter < max_iterations:
    previous_x = current_x
    current_x = previous_x - rate * slope(previous_x)
    y.append(value_y(current_x))
    x.append(current_x)
    delta_x = abs(previous_x - current_x)
    print(f"Iteration {iteration_counter+1}")
    iteration_counter += 1
    print(f"X = {current_x}")

print(f"Local Minima occurs at: {current_x}")
```

```
Iteration 1
```

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X = 1.9
Iteration 2
X = 1.8019999999999998
Iteration 3
X = 1.70596
Iteration 4
X = 1.6118408
Iteration 5
X = 1.519603984
Iteration 6
X = 1.42921190432
Iteration 7
X = 1.3406276662336
Iteration 8
X = 1.253815112908928
Iteration 9
X = 1.1687388106507495
Iteration 10
X = 1.0853640344377344
Iteration 11
X = 1.0036567537489798
Iteration 12
X = 0.9235836186740002
Iteration 13
X = 0.8451119463005202
Iteration 14
X = 0.7682097073745098
Iteration 15
X = 0.6928455132270196
Iteration 16
X = 0.6189886029624792
Iteration 17
X = 0.5466088309032295
Iteration 18
X = 0.4756766542851649
Iteration 19
X = 0.40616312119946163
Iteration 20
X = 0.3380398587754724
Iteration 21
X = 0.27127906159996296
Iteration 22
X = 0.2058534803679637
Iteration 23
X = 0.14173641076060442
Iteration 24
X = 0.07890168254539233
Iteration 25
```

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X = 0.017323648894484484
Iteration 26
X = -0.04302282408340521
Iteration 27
X = -0.10216236760173711
Iteration 28
X = -0.16011912024970237
Iteration 29
X = -0.21691673784470833
Iteration 30
X = -0.27257840308781417
Iteration 31
X = -0.3271268350260579
Iteration 32
X = -0.38058429832553675
Iteration 33
X = -0.432972612359026
Iteration 34
X = -0.4843131601118455
Iteration 35
X = -0.5346268969096086
Iteration 36
X = -0.5839343589714164
Iteration 37
X = -0.632255671791988
Iteration 38
X = -0.6796105583561483
Iteration 39
X = -0.7260183471890252
Iteration 40
X = -0.7714979802452447
Iteration 41
X = -0.8160680206403399
Iteration 42
X = -0.8597466602275331
Iteration 43
X = -0.9025517270229825
Iteration 44
X = -0.9445006924825228
Iteration 45
X = -0.9856106786328723
Iteration 46
X = -1.025898465060215
Iteration 47
X = -1.0653804957590107
Iteration 48
X = -1.1040728858438305
Iteration 49
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X = -1.1419914281269539
Iteration 50
X = -1.1791515995644148
Iteration 51
X = -1.2155685675731265
Iteration 52
X = -1.2512571962216639
Iteration 53
X = -1.2862320522972306
Iteration 54
X = -1.320507411251286
Iteration 55
X = -1.3540972630262602
Iteration 56
X = -1.387015317765735
Iteration 57
X = -1.4192750114104202
Iteration 58
X = -1.4508895111822118
Iteration 59
X = -1.4818717209585675
Iteration 60
X = -1.512234286539396
Iteration 61
X = -1.5419896008086083
Iteration 62
X = -1.571149808792436
Iteration 63
X = -1.5997268126165873
Iteration 64
X = -1.6277322763642557
Iteration 65
X = -1.6551776308369706
Iteration 66
X = -1.6820740782202313
Iteration 67
X = -1.7084325966558267
Iteration 68
X = -1.7342639447227102
Iteration 69
X = -1.759578665828256
Iteration 70
X = -1.784387092511691
Iteration 71
X = -1.808699350661457
Iteration 72
X = -1.832525363648228
Iteration 73
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X = -1.8558748563752634
Iteration 74
X = -1.8787573592477582
Iteration 75
X = -1.901182212062803
Iteration 76
X = -1.923158567821547
Iteration 77
X = -1.9446953964651161
Iteration 78
X = -1.9658014885358137
Iteration 79
X = -1.9864854587650975
Iteration 80
X = -2.0067557495897956
Iteration 81
X = -2.026620634598
Iteration 82
X = -2.04608822190604
Iteration 83
X = -2.065166457467919
Iteration 84
X = -2.0838631283185607
Iteration 85
X = -2.1021858657521895
Iteration 86
X = -2.1201421484371457
Iteration 87
X = -2.137739305468403
Iteration 88
X = -2.1549845193590347
Iteration 89
X = -2.171884828971854
Iteration 90
X = -2.188447132392417
Iteration 91
X = -2.204678189744569
Iteration 92
X = -2.2205846259496775
Iteration 93
X = -2.236172933430684
Iteration 94
X = -2.2514494747620706
Iteration 95
X = -2.266420485266829
Iteration 96
X = -2.2810920755614923
Iteration 97
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```
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Iteration 98
X = -2.3095608293692576
Iteration 99
X = -2.3233696127818724
Iteration 100
X = -2.336902220526235
Iteration 101
X = -2.3501641761157104
Iteration 102
X = -2.3631608925933962
Iteration 103
X = -2.375897674741528
Iteration 104
X = -2.3883797212466975
Iteration 105
X = -2.4006121268217635
Iteration 106
X = -2.4125998842853282
Iteration 107
X = -2.4243478865996217
Iteration 108
X = -2.4358609288676294
Iteration 109
X = -2.447143710290277
Iteration 110
X = -2.458200836084471
Iteration 111
X = -2.4690368193627816
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X = -2.479656082975526
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X = -2.4900629613160152
Iteration 114
X = -2.500261702089695
Iteration 115
X = -2.510256468047901
Iteration 116
X = -2.520051338686943
Iteration 117
X = -2.5296503119132043
Iteration 118
X = -2.53905730567494
Iteration 119
X = -2.5482761595614414
Iteration 120
X = -2.5573106363702127
Iteration 121
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Iteration 122
X = -2.5748411351699523
Iteration 123
X = -2.5833443124665534
Iteration 124
X = -2.591677426217222
Iteration 125
X = -2.599843877692878
Iteration 126
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Iteration 127
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Iteration 128
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Iteration 130
X = -2.638290559079748
Iteration 131
X = -2.6455247478981527
Iteration 132
X = -2.6526142529401895
Iteration 133
X = -2.6595619678813858
Iteration 134
X = -2.666370728523758
Iteration 135
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Iteration 136
X = -2.679582447674217
Iteration 137
X = -2.685990798720733
Iteration 138
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Iteration 139
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Iteration 141
X = -2.7103679107929732
Iteration 142
X = -2.7161605525771138
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X = -2.7218373415255717
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X = -2.72740059469506
Iteration 145
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Iteration 146
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Iteration 166
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X = -2.828712904560357
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Iteration 286
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Iteration 287
X = -2.984834604944367
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Iteration 289
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Iteration 290
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Iteration 291
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X = -2.9862916840175253
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Iteration 296
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Iteration 297
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X = -2.9906614479293494
Iteration 312
X = -2.9908482189707626
Iteration 313
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Iteration 314
X = -2.9912106294995207
Iteration 315
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Iteration 316
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Iteration 318
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X = -2.9942495032634273
Iteration 336
X = -2.9943645131981587
Iteration 337
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Iteration 338
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Iteration 358
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Iteration 359
X = -2.9964589571844593
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X = -2.9965297780407703
Iteration 361
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Iteration 362
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Iteration 370
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X = -2.9977295871131333
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X = -2.9977749953708708
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X = -2.997819495463453
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X = -2.9978631055541842
Iteration 385
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Iteration 386
X = -2.9979477265742385
Iteration 387
X = -2.9979887720427536
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X = -2.9980289966018985
Iteration 389
X = -2.9980684166698603
Iteration 390
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Iteration 391
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Iteration 392
X = -2.9981820092223392
Iteration 393
X = -2.9982183690378923
Iteration 394
X = -2.9982540016571346
Iteration 395
X = -2.998288921623992
Iteration 396
X = -2.998323143191512
Iteration 397
X = -2.998356680327682
Iteration 398
X = -2.9983895467211283
Iteration 399
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Iteration 400
X = -2.9984533206709716
Iteration 401
X = -2.9984842542575523
Iteration 402
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Iteration 403
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Iteration 404
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Iteration 405
X = -2.998601924388511
Iteration 406
X = -2.9986298859007405
Iteration 407
X = -2.998657288182726
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X = -2.998684142419071
Iteration 409
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Iteration 410
X = -2.9987362503792756
Iteration 411
X = -2.99876152537169
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X = -2.9987862948642565
Iteration 413
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Iteration 430
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Iteration 431
X = -2.999173184465324
Iteration 432
X = -2.999189720776018
Iteration 433
```

```
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Iteration 434
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Iteration 435
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X = -2.999501045986791
Iteration 457
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Iteration 459
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Iteration 480
X = -2.999692753929843
Iteration 481
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Iteration 483
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Iteration 504
X = -2.9998108039115277
Iteration 505
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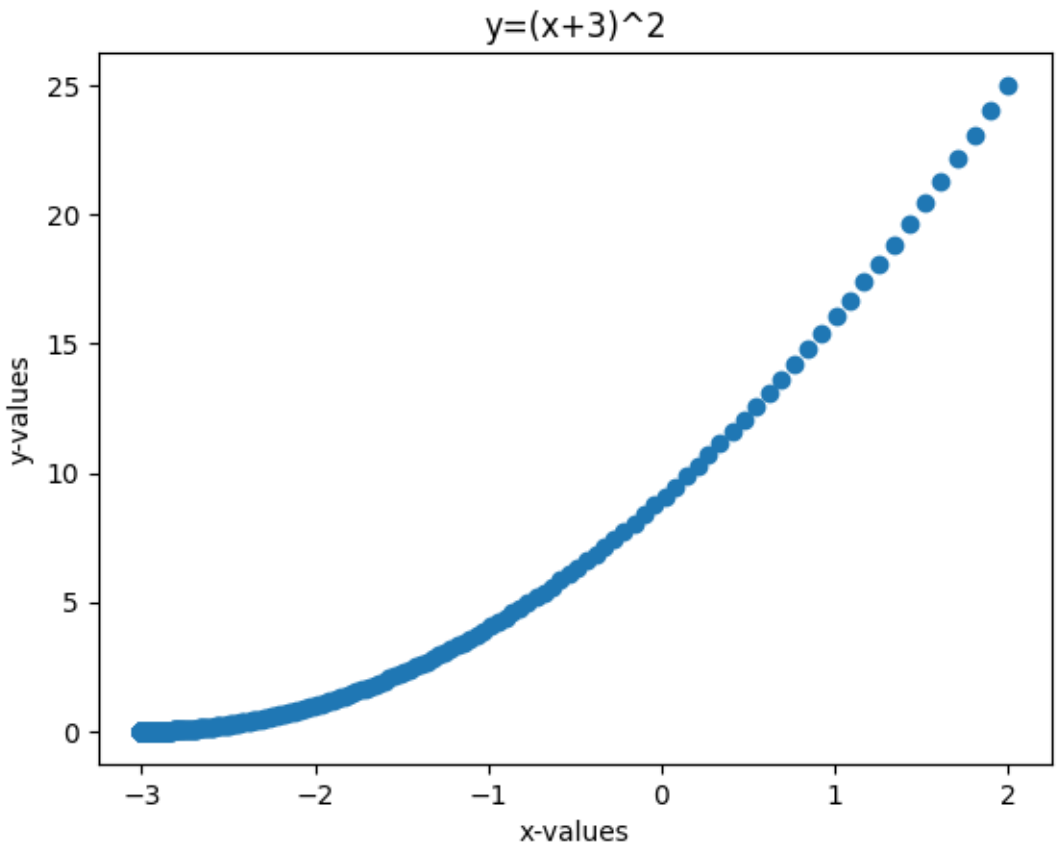
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Iteration 529
```

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Iteration 551
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Iteration 552
X = -2.999928259601195
Iteration 553
```

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X = -2.9999401866030424
Iteration 562
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Iteration 563
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Iteration 565
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X = -2.999945933426563
Iteration 567
X = -2.999947014758032
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X = -2.9999480744628713
Iteration 569
X = -2.999949112973614
Iteration 570
X = -2.999950130714142
Iteration 571
X = -2.999951128099859
Local Minima occurs at: -2.999951128099859
```

[4]:
```
import matplotlib.pyplot as plt
```

[5]:
```
plt.scatter(x,y)
plt.xlabel('x-values')
plt.ylabel('y-values')
plt.title('y=(x+3)^2')
plt.show()
```


y=(x+3)^2
25
20
15
10
5
0
y-values
−3
−2
−1
0
1
2
x-values

[ ]: